CO2_AT1 Scenario Based Question

Question-1:

Define a class named movieMagic with the following description: (20)

Instance variables/data members:
int year – to store the year of release of a movie
String title – to store the title of the movie.
float rating – to store the popularity rating of the movie.
(minimum rating = 0.0 and maximum rating = 5.0)

Member Methods:
(i) void accept() To input and store year, title and rating.
(ii) void display() To display the title of a movie and a message based on the rating as per the table below.

RATING	MESSAGE TO BE DISPLAYED
0.0 to 2.0	Flop
2.1 to 3.4	Semi-hit
3.5 to 4.5	Hit
4.6 to 5.0	Super Hit

Write a main method to create an object of the class and call the above member methods.

Question-2:

Define a class named BookFair with the following description: (20)

Instance variables/Data members:
String Bname – stores the name of the book.
double price – stores the price of the book.
Member Methods:

(i) void Input() – To input and store the name and the price of the book.
(ii) void calculate() – To calculate the price after discount. Discount is calculated based on the following criteria.

PRICE	DISCOUNT
Less than or equal to Rs 1000	2% of price
More than Rs 1000 and less than or equal to Rs 3000	10% of price
More than Rs 3000	15% of price

(iv)void display() – To display the name and price of the book after discount.

Write a main method to create an object of the class and call the above member methods

Question-3:

Define a class Electric Bill with the following specifications:

Instance Variable/ data member:
String n – to store the name of the customer
int units – to store the number of units consumed
double bill – to store the amount to paid

Member methods:
Void accept() – to accept the name of the customer and number of units consumed
Void calculate() – to calculate the bill as per the following tariff :
Number of units Rate per unit
First 100 units Rs.2.00
Next 200 units Rs.3.00
Above 300 units Rs.5.00

A surcharge of 2.5% charged if the number of units consumed is above 300 units.

Void print() – To print the details as follows :
Name of the customer ………
Number of units consumed ……
Bill amount …….

Write a main method to create an object of the class and call the above member methods.

Question-4:

Design a class RailwayTicket with the following description: (20)

Instance variables/data members:
String name: to store the name of the customer.
String coach: to store the type of coach customer wants to travel. long mobno: to store customer's mobile number.
int amt: to store basic amount of ticket.
int totalamt: to store the amount to be paid after updating the original amount.

Methods:
void accept(): to take input for name, coach, mobile number and amount.
void update(): to update the amount as per the coach selected. Extra amount to be added in the amount as follows:

Type of coaches	Amount
First_AC	700
Second_AC	500
Third_AC	250
sleeper	None

void display(): To display all details of a customer such as name, coach, total amount and mobile number.

Write a main() method to create an object of the class and call the above methods.

Question-5:

Design a class named ShowRoom with the following description: (20)

Instance variables/data members:
String name: to store the name of the customer.
long mobno: to store the mobile number of the customer.
double cost: to store the cost of the items purchased.
double dis: to store the discount amount.
double amount: to store the amount to be paid after discount.

Member methods:
void input(): to input customer name, mobile number, cost.
void calculate(): to calculate discount on the cost of purchased items, based on the following criteria:

COST	DISCOUNT (IN PERCENTAGE)
Less than or equal to Rs. 10000	5%
More than Rs. 10000 and less than or equal to Rs. 20000	10%
More than Rs. 20000 and less than or equal to Rs. 35000	15%
More than Rs. 35000	20%

void display(): to display customer name, mobile number, amount to be paid after discount.

Write a main() method to create an object of the class and call the above member methods.

CO3_AT1 Analytical Problem Solving

Question-1:

Define a class to represent a bank account. Include the following members:

Data members:
1. Name of the depositor
2. Account number
3. Type of account
4. Balance amount in the account

Member functions:
1. To assign initial values
2. To deposit an amount checking with account number and name of the customer
3. To withdraw an amount after checking the balance
4. To display name and balance

Write a main program to test the program use constructors and destructors.

Question-2:

Use Classes, Object, Constructor and complete the following application.

A book shop maintains the inventory of books that re being sold at the shop. The list includes details such as author, title, price, publisher and stock position. Whenever a customer wants a book, the sales person inputs the title and author and the system searches the list and displays whether it is available or not. If it is not, an appropriate message is displayed. If it is, then the system displays the book details and requests for the number of copies required. If the requested copies are available, the total cost of the requested copies is displayed; otherwise, the message “Required copies not in stock” is displayed. Design a system using a class called books with suitable member functions and constructor.

Also improve the system to incorporate the following features:

(i) The price of the books should be updated as and when required, Use a private member function to implement this.
(ii) The stock value of each book should be automatically updated as soon as a transaction is completed.
(iii) The number of successful and unsuccessful transactions should be recorded for the purpose of statistical analysis.

Question-3:

Create a class FLOAT that contains one float data member. Overload all four arithmetic operators (+,-,*,/) so that they operate on the objects of class FLOAT.

Question-4:

Define a class String. Use overloaded ==, >, < operators to compare two strings.

CO4_AT2 CASE Study

Question-1:

Write a C++ program to calculate the percentage of a student using multi-level inheritance. Accept the marks of three subjects in base class. A class will derived from the above mentioned class which includes a function to find the total marks obtained and another class derived from this class which calculates and displays the percentage of student.

Question-2:

Design a C++ program using hierarchical inheritance with a base class Person containing the data members name, address, and phone_no. Derive two classes, Employee and Manager, from the class Person. The Employee class should contain eno and ename, while the Manager class should contain designation, department_name, and basic_salary.

Write a menu-driven C++ program to:

a. Accept all details of n employees and n managers.
b. Display the details of all employees and managers.
c. Find and display the manager who has the highest basic salary.
d. Exit the program.

Question-3:

1. Consider two base classes
worker(int code, char name, float salary), officer(float DA, HRA)
class manger(float TA(is 10% of salary), gross sal) is derived from both base classes. Write necessary member function.

Question-4:

Write a C++ program to define a base class Item (item-no, name, price). Derive a class Discounted-Item (discount-percent). A customer purchases 'n' items. Display the item-wise bill and total amount using appropriate format.

CO2 AT1

Question 1: movieMagic

```
#include <iostream>
using namespace std;

class movieMagic {
    int year;
    string title;
    float rating;
public:
    void accept() {
        cout << "Enter title: "; cin >> title;
        cout << "Enter year: "; cin >> year;
        cout << "Enter rating (0-5): "; cin >> rating;
    }
    void display() {
        cout << "Title: " << title << endl;
        if (rating <= 2.0) cout << "Flop" << endl;
        else if (rating <= 3.4) cout << "Semi-hit" << endl;
        else if (rating <= 4.5) cout << "Hit" << endl;
        else cout << "Super Hit" << endl;
    }
};

int main() {
    movieMagic m;
    m.accept();
    m.display();
    return 0;
}
```

Question 2: BookFair

```
#include <iostream>
using namespace std;

class BookFair {
    string Bname;
    double price;
public:
    void Input() {
        cout << "Enter book name: "; cin >> Bname;
        cout << "Enter price: "; cin >> price;
    }
    void calculate() {
        if (price <= 1000) price -= price * 0.02;
        else if (price <= 3000) price -= price * 0.10;
        else price -= price * 0.15;
    }
    void display() {
```

```cpp
        cout << "Book: " << Bname << ", Price after discount: " << price << endl;
    }
};

int main() {
    BookFair b;
    b.Input();
    b.calculate();
    b.display();
    return 0;
}
```

Question 3: ElectricBill

```cpp
#include <iostream>
using namespace std;

class ElectricBill {
    string n;
    int units;
    double bill;
public:
    void accept() {
        cout << "Enter customer name: "; cin >> n;
        cout << "Enter units consumed: "; cin >> units;
    }
    void calculate() {
        if (units <= 100)
            bill = units * 2.0;
        else if (units <= 300)
            bill = 100 * 2.0 + (units - 100) * 3.0;
        else {
            bill = 100 * 2.0 + 200 * 3.0 + (units - 300) * 5.0;
            bill += bill * 0.025;
        }
    }
    void print() {
        cout << "Name of the customer: " << n << endl;
        cout << "Number of units consumed: " << units << endl;
        cout << "Bill amount: " << bill << endl;
    }
};

int main() {
    ElectricBill e;
    e.accept();
    e.calculate();
    e.print();
    return 0;
}
```

Question 4: RailwayTicket

```
#include <iostream>
using namespace std;

class RailwayTicket {
    string name;
    string coach;
    long mobno;
    int amt;
    int totalamt;
public:
    void accept() {
        cout << "Enter name: "; cin >> name;
        cout << "Enter coach type (First_AC/Second_AC/Third_AC/sleeper): "; cin >>
coach;
        cout << "Enter mobile number: "; cin >> mobno;
        cout << "Enter basic amount: "; cin >> amt;
    }
    void update() {
        if (coach == "First_AC") totalamt = amt + 700;
        else if (coach == "Second_AC") totalamt = amt + 500;
        else if (coach == "Third_AC") totalamt = amt + 250;
        else totalamt = amt;
    }
    void display() {
        cout << "Name: " << name << endl;
        cout << "Coach: " << coach << endl;
        cout << "Total Amount: " << totalamt << endl;
        cout << "Mobile Number: " << mobno << endl;
    }
};

int main() {
    RailwayTicket r;
    r.accept();
    r.update();
    r.display();
    return 0;
}
```

Question 5: ShowRoom

```
#include <iostream>
using namespace std;

class ShowRoom {
    string name;
    long mobno;
    double cost, dis, amount;
public:
    void input() {
```

```
        cout << "Enter name: "; cin >> name;
        cout << "Enter mobile number: "; cin >> mobno;
        cout << "Enter cost: "; cin >> cost;
    }
    void calculate() {
        if (cost <= 10000) dis = cost * 0.05;
        else if (cost <= 20000) dis = cost * 0.10;
        else if (cost <= 35000) dis = cost * 0.15;
        else dis = cost * 0.20;
        amount = cost - dis;
    }
    void display() {
        cout << "Name: " << name << endl;
        cout << "Mobile Number: " << mobno << endl;
        cout << "Amount after discount: " << amount << endl;
    }
};

int main() {
    ShowRoom s;
    s.input();
    s.calculate();
    s.display();
    return 0;
}
```

CO3 AT1

Question 1: BankAccount (constructors & destructors)

```
#include <iostream>
using namespace std;

class BankAccount {
    string name;
    long accNo;
    string accType;
    double balance;
public:
    BankAccount(string n, long a, string t, double b) {
        name = n; accNo = a; accType = t; balance = b;
        cout << "Account created for " << name << endl;
    }
    void deposit(long a, string n, double amt) {
        if (a == accNo && n == name) {
            balance += amt;
            cout << "Deposit successful. New balance: " << balance << endl;
        } else {
            cout << "Account details do not match." << endl;
        }
    }
```

```
    void withdraw(double amt) {
        if (amt <= balance) {
            balance -= amt;
            cout << "Withdrawal successful. New balance: " << balance << endl;
        } else {
            cout << "Insufficient balance." << endl;
        }
    }
    void display() {
        cout << "Name: " << name << ", Balance: " << balance << endl;
    }
    ~BankAccount() {
        cout << "Account of " << name << " closed." << endl;
    }
};

int main() {
    BankAccount acc("Ravi", 1001, "Savings", 5000);
    acc.deposit(1001, "Ravi", 2000);
    acc.withdraw(1500);
    acc.display();
    return 0;
}
```

Question 2: Book Shop Inventory

```
#include <iostream>
using namespace std;

class Books {
    string author, title, publisher;
    double price;
    int stock;
    static int successTrans, failTrans;
    void updatePrice(double newPrice) { price = newPrice; } // private member
function

public:
    Books(string a, string t, string p, double pr, int s) {
        author = a; title = t; publisher = p; price = pr; stock = s;
    }

    bool matches(string t, string a) {
        return (title == t && author == a);
    }

    void display() {
        cout << "Title: " << title << ", Author: " << author
             << ", Publisher: " << publisher << ", Price: " << price
             << ", Stock: " << stock << endl;
    }
```

```
    void sell(int copies) {
        if (copies <= stock) {
            cout << "Total cost: " << copies * price << endl;
            stock -= copies;
            successTrans++;
        } else {
            cout << "Required copies not in stock" << endl;
            failTrans++;
        }
    }

    void changePrice(double p) { updatePrice(p); }

    static void showStats() {
        cout << "Successful transactions: " << successTrans << endl;
        cout << "Unsuccessful transactions: " << failTrans << endl;
    }
};

int Books::successTrans = 0;
int Books::failTrans = 0;

int main() {
    Books b1("R.K. Narayan", "Malgudi Days", "Penguin", 250, 10);

    string t = "Malgudi Days", a = "R.K. Narayan";
    if (b1.matches(t, a)) {
        b1.display();
        int copies;
        cout << "Enter copies required: ";
        cin >> copies;
        b1.sell(copies);
    } else {
        cout << "Book not available" << endl;
    }

    b1.changePrice(275);
    Books::showStats();
    return 0;
}
```

Question 3: FLOAT operator overloading

```
#include <iostream>
using namespace std;

class FLOAT {
    float val;
public:
    FLOAT(float v = 0) { val = v; }
```

```
    FLOAT operator+(FLOAT f) { return FLOAT(val + f.val); }
    FLOAT operator-(FLOAT f) { return FLOAT(val - f.val); }
    FLOAT operator*(FLOAT f) { return FLOAT(val * f.val); }
    FLOAT operator/(FLOAT f) { return FLOAT(val / f.val); }

    void display() { cout << val << endl; }
};

int main() {
    FLOAT a(10.5), b(2.5);
    FLOAT sum = a + b;
    FLOAT diff = a - b;
    FLOAT prod = a * b;
    FLOAT quo = a / b;

    cout << "Sum: "; sum.display();
    cout << "Difference: "; diff.display();
    cout << "Product: "; prod.display();
    cout << "Quotient: "; quo.display();
    return 0;
}
```

Question 4: String comparison operator overloading

```
#include <iostream>
#include <cstring>
using namespace std;

class MyString {
    char str[100];
public:
    MyString(const char* s = "") { strcpy(str, s); }

    bool operator==(MyString s) { return strcmp(str, s.str) == 0; }
    bool operator>(MyString s) { return strcmp(str, s.str) > 0; }
    bool operator<(MyString s) { return strcmp(str, s.str) < 0; }
};

int main() {
    MyString s1("Apple"), s2("Banana");

    if (s1 == s2) cout << "Strings are equal" << endl;
    else if (s1 > s2) cout << "First string is greater" << endl;
    else if (s1 < s2) cout << "First string is smaller" << endl;

    return 0;
}
```

CO4 AT2

Question 1: Multi-level Inheritance - Percentage

```
#include <iostream>
using namespace std;

class Marks {
protected:
    float sub1, sub2, sub3;
public:
    void getMarks() {
        cout << "Enter marks of 3 subjects: ";
        cin >> sub1 >> sub2 >> sub3;
    }
};

class Total : public Marks {
protected:
    float total;
public:
    void calculateTotal() {
        total = sub1 + sub2 + sub3;
    }
};

class Percentage : public Total {
public:
    void calculatePercentage() {
        calculateTotal();
        float perc = total / 3;
        cout << "Total marks: " << total << endl;
        cout << "Percentage: " << perc << "%" << endl;
    }
};

int main() {
    Percentage p;
    p.getMarks();
    p.calculatePercentage();
    return 0;
}
```

Question 2: Hierarchical Inheritance - Person, Employee, Manager

```
#include <iostream>
using namespace std;

class Person {
protected:
    string name, address, phone_no;
public:
    void getPersonDetails() {
        cout << "Enter name: "; cin >> name;
        cout << "Enter address: "; cin >> address;
```

```
    cout << "Enter phone number: "; cin >> phone_no;
  }
  void showPersonDetails() {
    cout << "Name: " << name << ", Address: " << address << ", Phone: " <<
phone_no << endl;
  }
};

class Employee : public Person {
  int eno;
  string ename;
public:
  void getDetails() {
    getPersonDetails();
    cout << "Enter employee number: "; cin >> eno;
    cout << "Enter employee name: "; cin >> ename;
  }
  void showDetails() {
    showPersonDetails();
    cout << "Emp No: " << eno << ", Emp Name: " << ename << endl;
  }
};

class Manager : public Person {
  string designation, department_name;
  double basic_salary;
public:
  void getDetails() {
    getPersonDetails();
    cout << "Enter designation: "; cin >> designation;
    cout << "Enter department: "; cin >> department_name;
    cout << "Enter basic salary: "; cin >> basic_salary;
  }
  void showDetails() {
    showPersonDetails();
    cout << "Designation: " << designation << ", Department: " <<
department_name
        << ", Salary: " << basic_salary << endl;
  }
  double getSalary() { return basic_salary; }
  string getName() { return name; }
};

int main() {
  int n, choice;
  Employee* emp = nullptr;
  Manager* mgr = nullptr;

  do {
```

```
        cout << "\n1. Accept Employees & Managers\n2. Display All\n3. Highest Salary
Manager\n4. Exit\nChoice: ";
        cin >> choice;

        switch (choice) {
            case 1: {
                cout << "Enter number of employees: "; cin >> n;
                emp = new Employee[n];
                for (int i = 0; i < n; i++) emp[i].getDetails();

                cout << "Enter number of managers: "; cin >> n;
                mgr = new Manager[n];
                for (int i = 0; i < n; i++) mgr[i].getDetails();
                break;
            }
            case 2: {
                cout << "\nEmployees:\n";
                for (int i = 0; i < n; i++) emp[i].showDetails();
                cout << "\nManagers:\n";
                for (int i = 0; i < n; i++) mgr[i].showDetails();
                break;
            }
            case 3: {
                double maxSal = mgr[0].getSalary();
                string maxName = mgr[0].getName();
                for (int i = 1; i < n; i++) {
                    if (mgr[i].getSalary() > maxSal) {
                        maxSal = mgr[i].getSalary();
                        maxName = mgr[i].getName();
                    }
                }
                cout << "Highest paid manager: " << maxName << ", Salary: " << maxSal
<< endl;
                break;
            }
        }
    } while (choice != 4);

    delete[] emp;
    delete[] mgr;
    return 0;
}
```

Multiple Inheritance - Manager from Worker & Officer

```
#include <iostream>
using namespace std;

class Worker {
protected:
    int code;
```

```
    char name[30];
    float salary;
public:
    void getWorkerData() {
        cout << "Enter code: "; cin >> code;
        cout << "Enter name: "; cin >> name;
        cout << "Enter salary: "; cin >> salary;
    }
};

class Officer {
protected:
    float DA, HRA;
public:
    void getOfficerData() {
        cout << "Enter DA: "; cin >> DA;
        cout << "Enter HRA: "; cin >> HRA;
    }
};

class Manager : public Worker, public Officer {
    float TA, grossSal;
public:
    void getData() {
        getWorkerData();
        getOfficerData();
    }
    void calculate() {
        TA = 0.10 * salary;
        grossSal = salary + DA + HRA + TA;
    }
    void display() {
        cout << "Name: " << name << endl;
        cout << "TA: " << TA << endl;
        cout << "Gross Salary: " << grossSal << endl;
    }
};

int main() {
    Manager m;
    m.getData();
    m.calculate();
    m.display();
    return 0;
}
```

Question 4: Item, Discounted-Item - Billing

```
#include <iostream>
#include <iomanip>
using namespace std;
```

```
class Item {
protected:
    int itemNo;
    string name;
    double price;
public:
    void getItemData() {
        cout << "Enter item number: "; cin >> itemNo;
        cout << "Enter item name: "; cin >> name;
        cout << "Enter price: "; cin >> price;
    }
};

class DiscountedItem : public Item {
    double discountPercent;
    int qty;
public:
    void getDiscountData() {
        getItemData();
        cout << "Enter discount percent: "; cin >> discountPercent;
        cout << "Enter quantity: "; cin >> qty;
    }
    double getBillAmount() {
        double discPrice = price - (price * discountPercent / 100);
        return discPrice * qty;
    }
    void displayBill() {
        double amt = getBillAmount();
        cout << left << setw(8) << itemNo << setw(15) << name
             << setw(10) << price << setw(6) << qty
             << setw(10) << amt << endl;
    }
};

int main() {
    int n;
    cout << "Enter number of items: ";
    cin >> n;

    DiscountedItem* items = new DiscountedItem[n];
    double total = 0;

    for (int i = 0; i < n; i++)
        items[i].getDiscountData();

    cout << left << setw(8) << "ItemNo" << setw(15) << "Name"
         << setw(10) << "Price" << setw(6) << "Qty" << setw(10) << "Amount" << endl;

    for (int i = 0; i < n; i++) {
```

```
        items[i].displayBill();
        total += items[i].getBillAmount();
    }

    cout << "Total Amount: " << total << endl;

    delete[] items;
    return 0;
}
```